

sipa-voice-gate

The voice agent that checks itself before it acts.

PROBLEM

Voice agents that act blindly

Voice agents that can spend money, delete files, or send messages with no confirmation step are a live safety gap.

Every other entry at this hackathon builds a faster, smoother voice assistant. None of them stop to check what they're about to do before doing it.

SOLUTION

A stricter loop, not a bigger model

Before any action with real consequences — send money, delete a file, email someone, make a purchase — the agent speaks back a structured confirmation of what it's about to do and the chain of consequences, then waits for an explicit spoken "yes."

Every action, taken or blocked, is logged with a verifiable, hash-chained receipt.

HOW IT WORKS

Six steps

1. AssemblyAI Universal-Streaming STT transcribes in real time
2. AssemblyAI PII redaction strips secrets before the LLM sees them
3. AssemblyAI LeMUR extracts a structured action from the transcript
4. Consequence-gate classifies it: reversible? moves money? deletes?
5. Low consequence -> acts + logs a receipt. High -> speaks the consequence chain and waits for an explicit "yes"
6. Every action writes a hash-chained receipt; a verifier catches tampering

TECH STACK

Real AssemblyAI, real gate, real receipts

AssemblyAI: Universal-Streaming STT, PII redaction, LeMUR

ElevenLabs: TTS for the agent's spoken confirmations

Python: consequence-gate, hash-chained receipt log, verifier

21 tests green, runs the full loop with zero API keys (run_demo.py)

TEAM — SIPAOS

Three people, three tracks

Aelin AquaSoul — consequence-gate design, core package, receipts

+ verifier, real PII redaction + ElevenLabs TTS wiring, demo UI

Gephel — voice pipeline (AssemblyAI STT integration, agent wiring)

Benjamin Hong — security review of the consequence-gate,
adversarial test suite

***Not a bigger model —
a stricter loop around it.***